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Clinical Trial Biomarker Analytics and Advanced Data Visualization Pipelines

The project began with exploratory analysis of the dataset to better understand how biomarkers are represented in clinical research. Because the Excel file contained multiple worksheets, the data was first inspected across sheets to determine which dataset should be used for analysis. Initial exploratory visualizations were created using the CTO worksheet to verify that the data could be successfully loaded and visualized. These preliminary charts examined the distribution of intervention classes and therapeutic classes across outcome categories. The results showed that drug-based interventions dominated across all outcome categories, while biological vaccine interventions represented a smaller proportion. Additionally, the DTT therapeutic class accounted for the majority of observations, indicating that the dataset was strongly focused on dementia-related therapeutic targets.

During this exploratory stage, it was verified that the CTO worksheet was not the dataset intended for visualization analysis. After reviewing the dataset structure and project guidance, the analysis shifted to the “Data for Charts — Amanda Edits” worksheet, which contained the cleaned and organized data prepared specifically for visualization. All subsequent analysis and visualizations were therefore developed using this dataset.

The analysis was conducted in Python using Google Colab, primarily utilizing Pandas for data manipulation and Matplotlib, Seaborn, and Plotly Express for visualization. One technical challenge encountered early in the process involved accessing the correct worksheet from the Excel file. Initially, only the first sheet was being loaded, which resulted in blank graphs when attempting to visualize the data. This issue was resolved by loading all worksheets using the ‘sheet_name=None’ parameter and selecting the appropriate sheet for analysis.

Several visualizations were then developed to examine patterns in biomarker usage. These included stacked bar charts showing how biomarkers are used across outcome categories, visualizations of biomarker types across clinical trial phases, hierarchical sunburst charts displaying relationships between therapeutic classes, biomarker types, and individual biomarkers, and heatmaps illustrating measurement methods across therapeutic classes. The results revealed that fluid biomarkers dominate early clinical trial phases, while imaging biomarkers become more prominent in later phases. In addition, certain measurement methods, such as cerebrospinal fluid (CSF), appeared frequently within specific therapeutic classes.

Overall, this work established a foundation for understanding the structure and distribution of biomarkers within the dataset. Future work will focus on expanding the interactive visualizations, refining the heatmap analysis, and exploring additional methods for representing high-dimensional biomarker data in a clear and interpretable way.